MIL-C-25/4D
21 June 1968
SUPERSEDING
MIL-C-25/4C
8 February 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLES CP53, CP54 AND CP55

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions: See figure 1 and tables I and II.

Case: Magnetic or nonmagnetic material, uninsulated.

Design and construction: Capacitors shall be of the design, construction and physical dimensions as specified in figure 1 and tables I and II.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC voltage rating: See table I.

Operating temperature range:
Characteristics E and F - -55 to +85° C.
Characteristic K - -55 to +125° C.

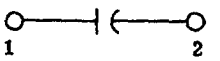
Vibration grade: 1 (10 to 55 Hz, incl.)

Terminal:

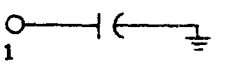
Symbol B - - - - Solder lug (nonremovable).

Circuit diagram:

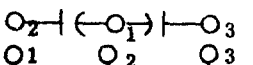
Symbol 1 - - - -



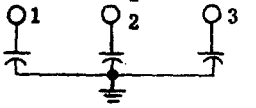
Symbol 2 - - - -



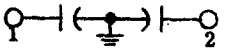
Symbol 4 - - - -



Symbol 5 - - - -



Symbol 6 - - - -



Voltage derating with temperature: See figure 2.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal) - As specified in table IV.
Capacitors shall withstand specified voltages with no permanent damage or open- or short-circuiting.

Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table V.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C (characteristics E and F):
Terminal to terminal - As specified in table V.
At 125° C (characteristic K):
Terminal to terminal - As specified in table V.

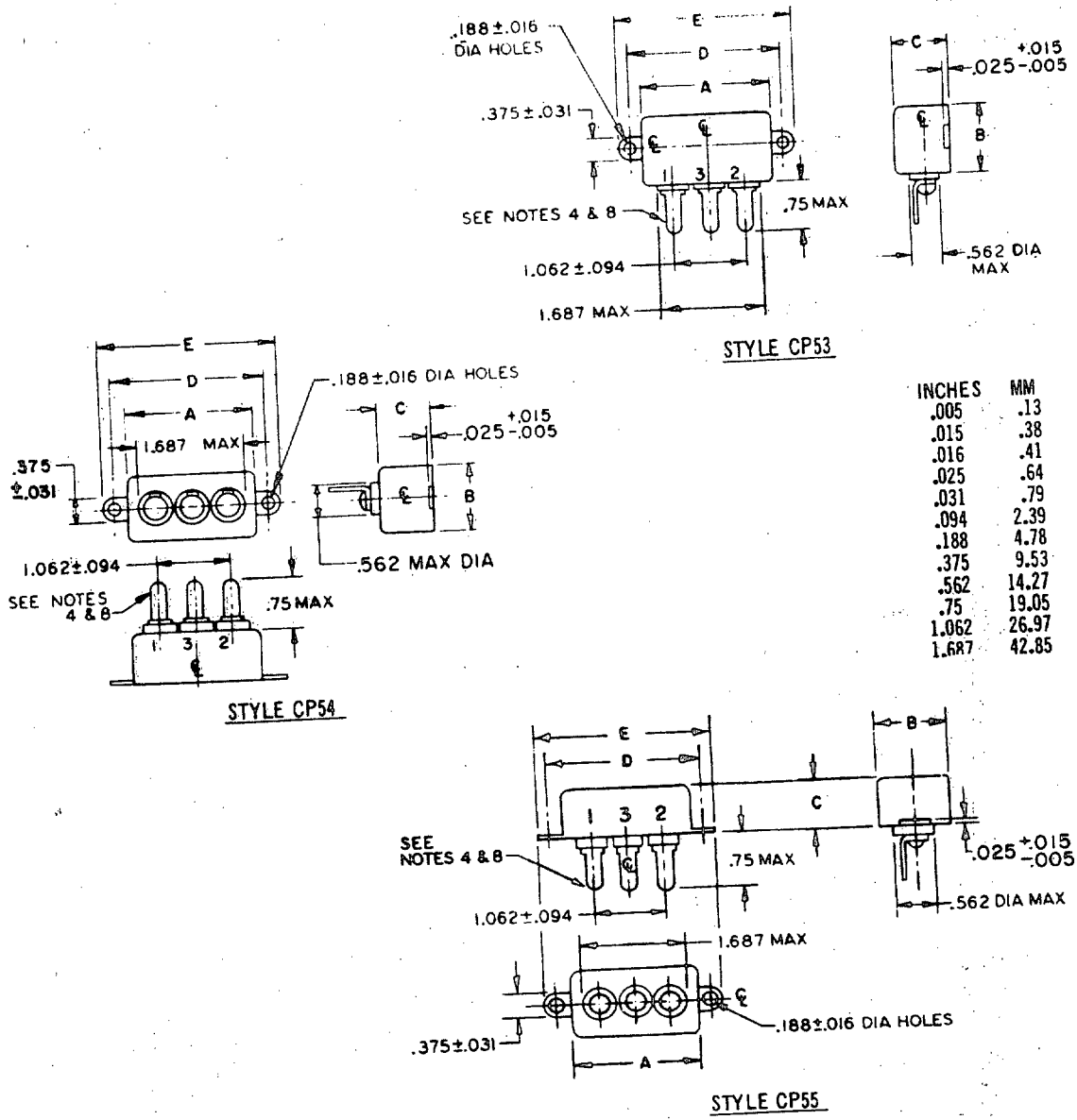
Capacitance: Method 305 of MIL-STD-202.
Measured capacitance shall be within the tolerance specified in table I.

Dissipation factor: When measured as specified in MIL-C-25, shall not exceed 1 percent.

Barometric pressure (flashover):
Method 105 of MIL-STD-202.
Test condition - B (50,000 feet).
Test potential - 125 percent of rated dc voltage, applied for a minimum of 1 minute, without visible damage, external flashover, or open- or short-circuiting.

FSC 5910

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INCHES	MM
.005	.13
.015	.38
.016	.41
.025	.64
.031	.79
.094	2.39
.188	4.78
.375	9.53
.562	14.27
.75	19.05
1.062	26.97
1.687	42.85

NOTES:
1. See tables I and II for additional dimensions.
2. All dimensions in inches.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Terminals need not be identified by numbers as shown. Shape of terminals and terminal lugs (drilled, punched, forked, or slotted) optional.
5. Mounting holes may be elongated.
6. For circuit 1, use terminals 1 and 2. For circuit 2, use terminal 1. For circuit 4, use terminal 1 for common connection and identify common terminal by the letter C on case adjacent to terminal. For circuit 6, use terminals 1 and 2.
7. Capacitors differ only in location of terminals. Style CP53, having the terminals on one side, contains certain values of capacitance in a smaller case than is possible with styles CP54 and CP55, the terminals of which are on top and bottom faces, respectively.
8. Terminal lugs shall be oriented as shown with respect to side of case bearing terminal identification numbers.

FIGURE 1. Styles CP53, CP54, and CP55 capacitors.

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TABLE I. Styles CP53, CP54 and CP55 characteristics and dimensions.

Type designation 1/	Dc voltage rating (see fig. 2)	Capacitance	Capacitance tolerance 2/	Characteristics	Vibration grade	Case size		
						Characteristic E	Characteristic F	Characteristic K
	Volts	µf						
CP5-B--B105K1	100	1	K	E, F	1	A2	A1	---
CP5-B-EB205K1	100	2	K	E	1	B1	---	---
CP5-B--B405K1	100	4	K	E, F	1	C2	C1	---
CP5-B-FB805K1	100	8	K	F	1	---	C1	---
CP5-B-FB106K1	100	10	K	F	1	---	C2	---
CP5-B-FC504K1	200	0.5	K	F	1	---	A1	---
CP5-B-EC105K1	200	1	K	E	1	B1	---	---
CP5-B-EC205K1	200	2	K	E	1	C1	---	---
CP5-B-KE503K1	400	0.05	K	K	1	---	---	A1
CP5-B-KE104K1	400	0.10	K	K	1	---	---	A1
CP5-B-KE254K1	400	0.25	K	K	1	---	---	A1
CP5-B-KE504K1	400	0.5	K	K	1	---	---	A2
CP5-B-KE105K1	400	1	K	K	1	---	---	B1
CP5-B-EF254K1	600	0.25	K	E	1	A1	---	---
CP5-B--F504K1	600	0.5	K	E, F	1	A3	A2	---
CP5-B--F105K1	600	1	K	E, F	1	B2	B1	---
CP5-B-EF205K1	600	2	K	E	1	C2 3/	---	---
CP5-B-EG503K1	1,000	0.05	K	E	1	A1	---	---
CP5-B-EG104K1	1,000	0.1	K	E	1	A1	---	---
CP5-B--G254K1	1,000	0.25	K	E, F	1	A2	A1	---
CP5-B-EG504K1	1,000	0.5	K	E	1	B1	---	---
CP5-B-EG105K1	1,000	1	K	E	1	C2 3/	---	---
CP5-B-EF503V1	600	0.05-0.05 4/	V	E	1	A1	---	---
CP5-B-EF104V1	600	0.1 -0.1	V	E	1	A1	---	---
CP5-B--F254V1	600	0.25-0.25	V	E, F	1	A3	A2	---
CP5-B--F504V1	600	0.5 -0.5	V	E, F	1	B2	B1	---
CP5-B-EF105V1	600	1 -1	V	E	1	C2 3/	---	---
CP5-B-EG503V1	1,000	0.05-0.05	V	E	1	A1	---	---
CP5-B--G104V1	1,000	0.1 -0.1	V	E, F	1	A2	A1	---
CP5-B-EG254V1	1,000	0.25-0.25	V	E	1	B1	---	---
CP5-B-EG504V1	1,000	0.5 -0.5	V	E	1	C2 3/	---	---
CP5-B5-E504V1	400	0.5 -0.5 -0.5	V	E, F	1	C2	C1	---
CP5-B5-F104V1	600	0.1 -0.1 -0.1	V	E, F	1	A2	A1	---
CP5-B5EF254V1	600	0.25-0.25-0.25	V	E	1	B1	---	---
CP5-B5EF504V1	600	0.5 -0.5 -0.5	V	E	1	C2 3/	---	---
CP5-B5EG503V1	1,000	0.05-0.05-0.05	V	E	1	A1	---	---
CP5-B5EG104V1	1,000	0.1 -0.1 -0.1	V	E	1	B1	---	---
CP5-B5EG254V1	1,000	0.25-0.25-0.25	V	E	1	C2 3/	---	---

1/ Complete type designation shall include additional symbols to complete style designation, indicate circuit (where applicable) and characteristics (where applicable).
2/ Capacitance tolerance in percent: K, ± 10; V. + 20, - 10.
3/ For styles CP54 and CP55, case size is C3.
4/ For multiple-unit capacitors, the watt-second rating is the sum of the watt-second ratings of the component sections.

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TABLE II. Case dimensions.

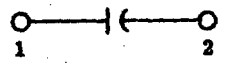
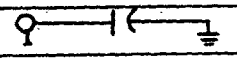
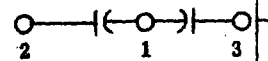
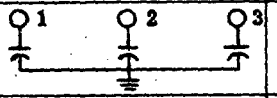
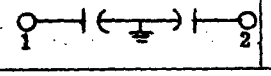
Case size	Dimensions				
	A ±0.062	B ±0.062	C $^{+0.062}_{-0.125}$	D ±0.031	E ±0.062
A1	1.812	1.000	0.750	2.125	2.500
A2	1.812	1.000	0.875	2.125	2.500
A3	1.812	1.000	1.000	2.125	2.500
B1	2.000	1.750	0.875	2.375	2.750
B2	2.000	1.750	1.000	2.375	2.750
C1	2.000	2.000	1.000	2.375	2.750
C2	2.000	2.000	1.125	2.375	2.750
C3	2.000	2.000	1.250	2.375	2.750

TABLE III. Millimeter equivalents of decimal inches.

In	mm	In	mm	In	mm
0.031	0.79	1.000	25.40	2.000	50.80
0.062	1.57	1.125	28.58	2.125	53.98
0.125	3.18	1.250	31.75	2.375	60.33
0.750	19.05	1.750	44.45	2.500	63.50
0.875	22.23	1.812	46.02	2.750	69.85

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TABLE IV. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage)	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 1/	1 2/
		Terminal to case	1 and 2 to case	400 5/	1 3/ 4/
				200 + 1,000 6/ volts	1 3/ 4/
2		Terminal to terminal	1 to case	200 1/	1 2/
4		Terminal to terminal	1 to 2 and 3 together	200 1/	1 7/
			2 to 3	200	1 7/
		Terminal to case	1, 2, and 3 to case	400	1 3/ 4/
				200 + 1,000 volts	1 3/ 4/
5		Terminal to terminal	1, 2, and 3 to case	200	1 7/
			1 and 2 to 3	200	1 7/
			1 to 2	200	1 7/
6		Terminal to terminal	1 and 2 to case	200	1 7/
			1 to 2	200	1 7/

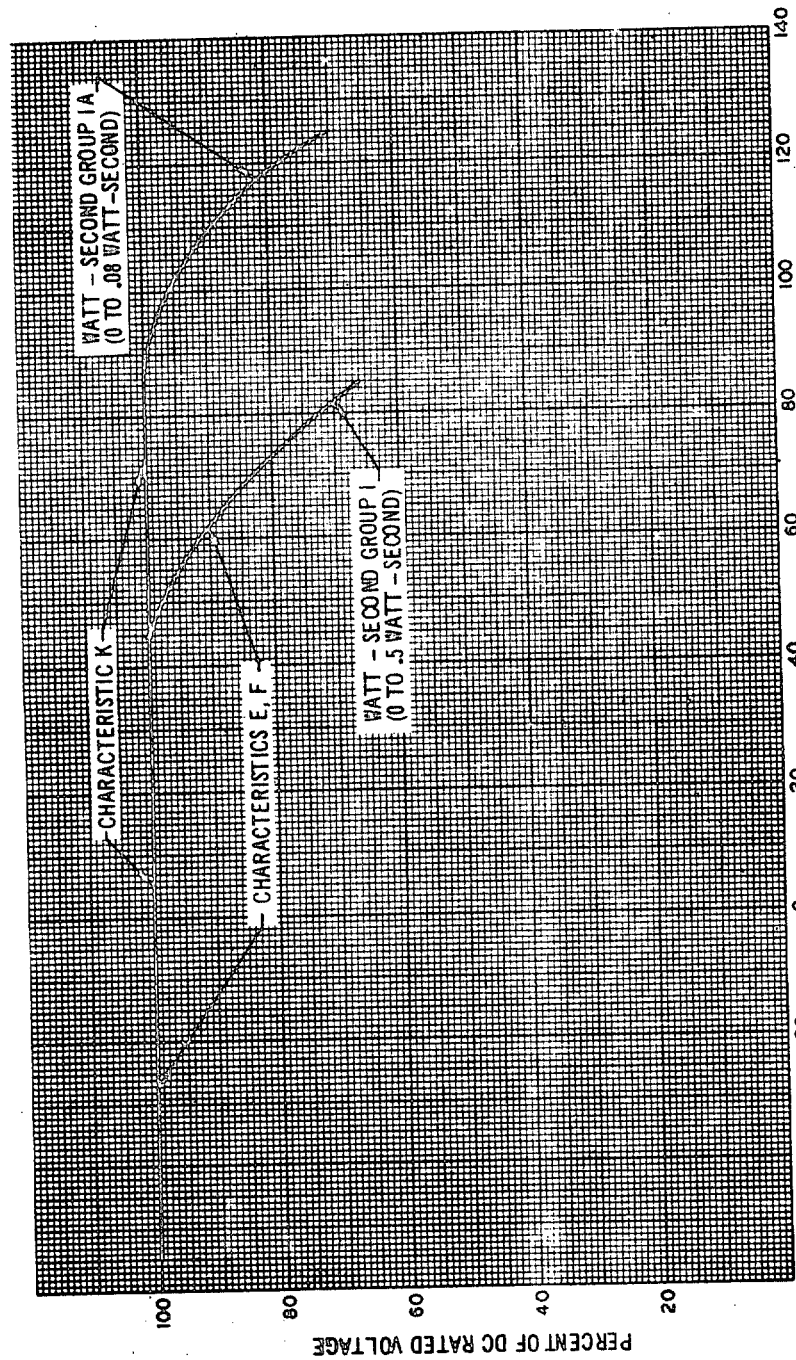
1/ 175 percent of rated dc voltage after temperature-and immersion-cycling and moisture resistance tests.
2/ For quality conformance inspection and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds or 250 percent of rated dc voltage for not less than 1 second.
3/ For quality conformance inspection, the period of test-voltage application shall be 1 second.
4/ For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.
5/ For capacitors having dc voltage ratings of 600 volts and less.
6/ For capacitors having dc voltage ratings greater than 600 volts.
7/ For quality conformance inspection, the application may be made for 1 second if the capacitor has passed group A inspection.

TABLE V. Terminal to terminal insulation resistance measurements.

Capacitance values	Minimum insulation resistance 1/		
	Characteristic E	Characteristic F	Characteristic K
	At 25° C	At 25° C	At 25° C
0.33 microfarads and less	6,000 megohms	4,500 megohms	18,000 megohms
greater than 0.33 microfarads	2,000 megohm-microfarads 2/	1,500 megohm-microfarads 2/	6,000 megohm-microfarads 2/
	At 85° C	At 85° C	
greater than 0.033 microfarads	20 megohm-microfarads 2/	15 megohm-microfarads 2/	---
			At 125° C
0.067 microfarads and less	---	---	150 megohms
greater than 0.067 microfarads	---	---	10 megohm-microfarads 2/

1/ A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.
2/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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NOTE: Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltages lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time.

FIGURE 2. Voltage derating for ambient temperature (characteristics E, F, and K).

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Vibration (grade 1, 10 to 55 Hz, incl):
Method 201 of MIL-STD-202.
Duration - 3 hours (60 minutes in each direction).
Measurements - An electrical signal of 1±0.2 kHz, at a level of 1±0.5 volts applied during the last 30 minutes in each direction. There shall be no momentary arcing or breakdowns, open- or short-circuiting, or mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion or obliteration of marking.

Temperature cycling: Method 102 of MIL-STD-202, test condition D, except that the test temperature of step 3 shall be as follows:
For characteristics E and F: 85° C.
For characteristic K: 125° C.

Immersion: Method 104 of MIL-STD-202, test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage, at 25° C:
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal)- As specified in table IV.
Insulation resistance at 25° C:
Terminal to terminal - Not less than 30 percent of the value specified in table V.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and E.
Test condition A (pull): Applied force - 5 pounds.
Test condition E (torque): 1 pound-inch.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal: Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be as follows:
For characteristics E and F: $+85^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
For characteristic K: $+125^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
Liquid-filled capacitors - 10 minutes on each of two opposite sides at test temperatures as follows:
For characteristics E and F: $+85^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
For characteristic K: $+125^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
There shall be no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature -
Conditions: -55° ± 3° C for 48 ± 4 hours with rated dc voltage applied.

Capacitance change with temperature:
Characteristic E - Not to exceed ±10 percent.
Characteristic F - Not to exceed +10, -30 percent.
Characteristic K - Not to exceed +7.5, -10 percent.

Life: Method 108 of MIL-STD-202, test condition B ($250\ ^{+8}_{-0}$ hours).
Electrical load: 120 percent of rated dc voltage.
Final measurements (at 25° C):
Insulation resistance:
Terminal to terminal: Not less than 30 percent of the value specified in table V.
Capacitance:
Characteristics E and F - Within 10 percent of initial value.
Characteristic K - Within 5 percent of initial value.
Dissipation factor:
Characteristics E and K - Change not more than 1 percent from initial value.
Characteristic F - Change not more than 1.5 percent from initial value.

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Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 11

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